

NCG WAREHOUSE SOLUTIONS

Asset Management

Executive Operations Manual

Comprehensive Documentation
ISO 55001 | IAS 16 | IFRS 16 Compliant

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1. Introduction and Purpose

This document serves as the comprehensive reference manual for the Asset Management module within the NCG Warehouse Solutions platform. It is designed for operations managers, warehouse supervisors, finance officers, procurement staff, and system administrators who interact with enterprise assets on a daily basis.

The Asset Management module provides end-to-end lifecycle management for all physical and intangible assets within the organisation. From the moment an asset is conceived as a requirement through to its eventual disposal or write-off, this system provides full traceability, financial accountability, and regulatory compliance.

This manual follows international standards including ISO 55001 (Asset Management Systems), IAS 16 (Property, Plant and Equipment), and IFRS 16 (Leases). Every workflow, calculation, and reporting function has been designed to meet these standards while remaining practical and user-friendly for daily operations.

The system integrates tightly with the broader NCG Warehouse Solutions ecosystem, including procurement, inventory management, finance and general ledger, and human resources modules. This integration ensures that asset-related transactions flow seamlessly across departmental boundaries without manual re-entry or reconciliation gaps.

1.1 Scope of This Document

This document covers all features, processes, and configurations related to asset management within the NCG Warehouse Solutions platform. It includes detailed process maps rendered as Mermaid diagrams, step-by-step operational procedures, field reference tables, role-permission matrices, and troubleshooting guides.

The intended audience includes department heads, asset custodians, finance teams responsible for depreciation and reporting, IT administrators managing system configuration, and auditors reviewing compliance with organisational policies and international accounting standards.

1.2 Document Conventions

Key Concept

Throughout this document, callout boxes like this one highlight important concepts, warnings, and best practice recommendations that readers should pay special attention to.

Screen descriptions use the heading 'What You Will See' followed by a detailed walkthrough of the user interface elements. Step-by-step procedures use numbered lists with clear action verbs. Tables provide structured reference data for fields, statuses, and permissions.

1.3 System Architecture Overview

The Asset Management module is built on a modern cloud-native architecture using React 18 for the frontend, Supabase (PostgreSQL) for the backend database, and Row-Level Security (RLS) for multi-tenant data isolation. The system supports multiple companies within a single deployment, with each company's data strictly segregated at the database level.

Edge functions handle server-side business logic such as depreciation batch processing, QR code generation, and notification dispatch. The CI/CD pipeline ensures that all changes are tested, validated, and deployed through a controlled release process.

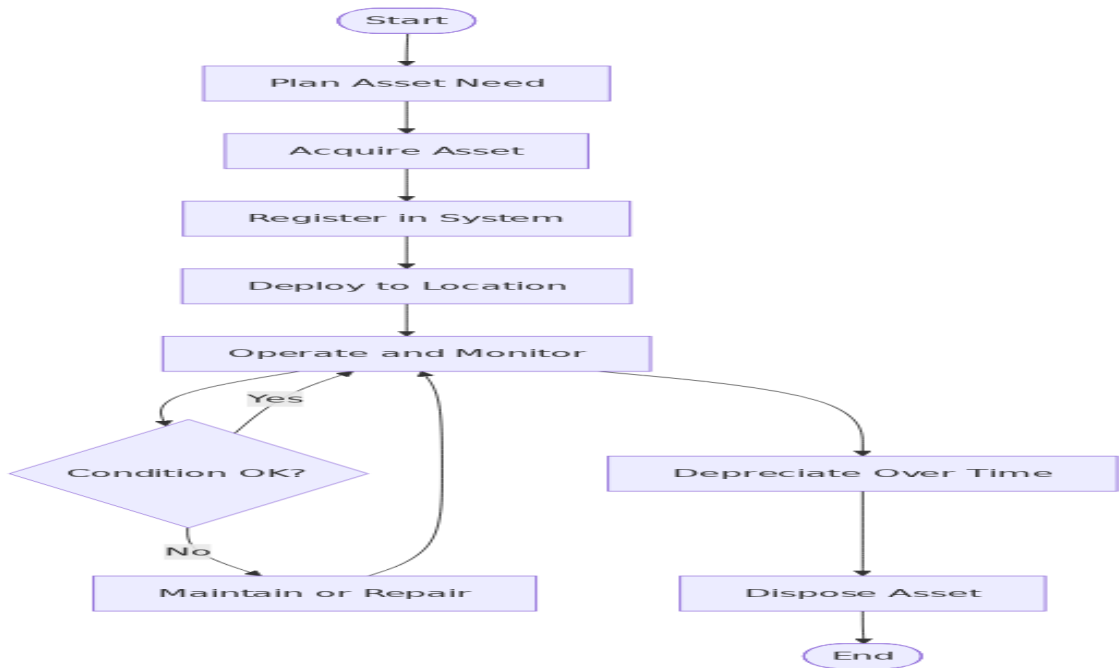


Figure 1: Asset Management System Architecture

1.4 Regulatory Compliance Framework

Standard	Scope	How We Comply
ISO 55001	Asset Management Systems	Full lifecycle tracking, risk-based decision making, performance monitoring
IAS 16	Property, Plant & Equipment	Cost model, revaluation model, component depreciation, impairment testing
IFRS 16	Leases	Right-of-use asset recognition, lease liability measurement, modification handling
SOX Section 404	Internal Controls	Segregation of duties, approval workflows, audit trails, access controls
GAAP	Generally Accepted Principles	Consistent depreciation methods, materiality thresholds, disclosure requirements

2. Asset Master Management

The Asset Master is the central registry of all asset types within the organisation. Unlike individual physical assets (which are tracked in the warehouse assets table), the Asset Master defines the template or blueprint for each class of asset. Think of it as a product catalogue for assets: it defines the standard name, brand, category, expected purchase price, depreciation parameters, and other attributes that apply to all instances of that asset type.

When a new laptop model is approved for use across the organisation, for example, a single Asset Master record is created with the model name, brand, standard configuration, expected price, useful life, and depreciation method. When individual laptops are purchased and received, each one becomes a warehouse asset linked back to this master record.

This two-tier architecture (master template plus individual instances) provides significant operational benefits. It ensures consistency in how similar assets are classified, depreciated, and reported. It simplifies procurement by providing a pre-approved catalogue of asset types. And it enables aggregate reporting across all instances of a particular asset type.

2.1 What You Will See

The Asset Master screen is accessed from the main navigation under Warehouse > Asset Management > Asset Master. The screen displays a searchable, sortable table of all registered asset masters with columns for asset name, brand, category, purchase price, current value, status, and creation date. A prominent 'Add Asset Master' button appears in the top-right corner.

Each row in the table includes action buttons for viewing details, editing the record, and accessing the purchase history. The table supports pagination, column sorting, and a global search field that filters across asset name and brand fields simultaneously.

2.2 Creating an Asset Master Record

2.2.1 Step-by-Step Procedure

- Navigate to Warehouse > Asset Management > Asset Master
- Click the 'Add Asset Master' button in the top-right corner
- Complete the required fields: Asset Name (mandatory), Brand, Category, and Subcategory
- Enter financial parameters: Purchase Price, Current Value, Depreciation Method, Useful Life, Salvage Value
- Optionally upload an image of the asset type for visual identification
- Add a description providing any additional context about the asset type
- Click 'Save' to create the record

2.2.2 Field Reference

Field	Type	Required	Description
Asset Name	Text	Yes	Descriptive name for the asset type (e.g., Dell Latitude 5540 Laptop)

Brand	Text	No	Manufacturer or brand name
Category	Dropdown	No	Level 1 asset category from the category hierarchy
Subcategory	Dropdown	No	Level 2 asset category, filtered by selected parent category
Purchase Price	Currency	No	Standard or expected purchase price for this asset type
Current Value	Currency	No	Current estimated market or book value
Depreciation Method	Dropdown	No	Straight-Line or Declining Balance
Depreciation Rate	Percentage	No	Annual depreciation rate (for declining balance method)
Useful Life (Years)	Number	No	Expected useful life in years (for straight-line method)
Salvage Value	Currency	No	Estimated residual value at end of useful life
Purchase Date	Date	No	Date of original or most recent purchase
Image	File Upload	No	Photo or diagram of the asset type
Description	Text Area	No	Free-text description with additional details
Status	Toggle	No	Active or Inactive (defaults to Active)

2.3 Purchase History Tracking

Each Asset Master record maintains a complete purchase history log. Every time the organisation purchases additional units of this asset type, a purchase history entry should be created. This provides a complete audit trail of procurement activity and enables trend analysis of purchase prices over time.

The purchase history includes the purchase date, unit price at time of purchase, vendor name, quantity purchased, and any notes. This data feeds into the procurement analytics dashboard and helps identify opportunities for volume discounts or vendor consolidation.

2.4 Best Practices

ISO 55001 Recommendation

Maintain a standardised naming convention for Asset Master records. Use the format: [Brand] [Model] [Specification]. For example: 'Dell Latitude 5540 i7/16GB/512GB'. This ensures consistency and makes searching and reporting more effective.

- Review Asset Master records quarterly to deactivate obsolete or discontinued asset types
- Ensure depreciation parameters are reviewed and approved by the finance team before creation
- Upload clear, high-resolution images to aid in physical identification during audits
- Use the description field to record any special handling, compliance, or safety requirements

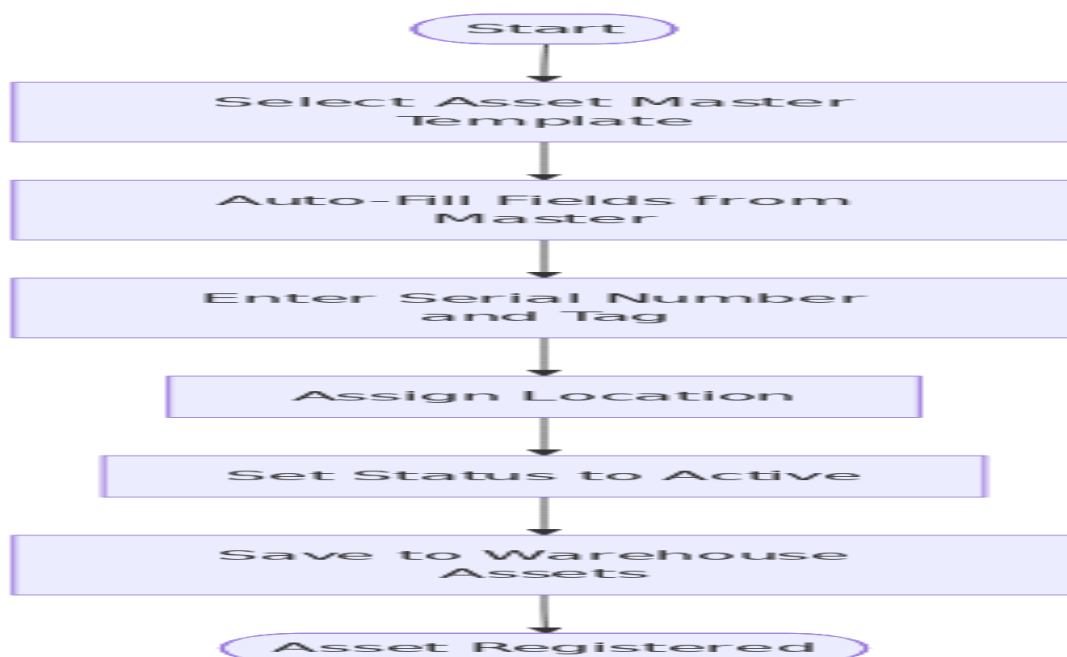


Figure 2: Asset Master Data Flow

3. Asset Category Hierarchy

The asset category system provides a hierarchical classification structure for organising assets into logical groups. The system supports a two-level hierarchy: Level 1 categories represent broad asset classes (such as IT Equipment, Furniture, Vehicles, or Building Infrastructure), while Level 2 subcategories provide more specific classification within each class.

A well-designed category hierarchy is essential for meaningful financial reporting, regulatory compliance, and operational analysis. Categories determine which general ledger accounts are used for asset-related journal entries, which depreciation policies apply by default, and how assets are grouped in management reports.

The category hierarchy also drives the automatic item code generation system. Each Level 1 category is assigned a three-letter mnemonic code (e.g., ITE for IT Equipment, FUR for Furniture, VEH for Vehicles), which forms the middle segment of the INV-{CAT}-XXX item code format used throughout the warehouse module.

3.1 What You Will See

The Category Management screen displays a tree view of all categories with parent-child relationships clearly indicated through indentation. Each category shows its name, description, number of associated assets, and action buttons for editing or adding subcategories. A search field at the top filters the tree in real time.

3.2 Category Design Guidelines

Best Practice

Limit Level 1 categories to 8-12 major asset classes. Too many top-level categories fragment reporting; too few create overly broad groups that lack analytical value. The subcategory level should provide the specificity needed for operational management.

Level 1 Category	Code	Example Subcategories
IT Equipment	ITE	Laptops, Desktops, Servers, Networking, Peripherals
Furniture	FUR	Office Desks, Chairs, Filing Cabinets, Conference Tables
Vehicles	VEH	Sedans, SUVs, Trucks, Forklifts, Trailers
Building Infrastructure	BLD	HVAC Systems, Electrical, Plumbing, Security Systems
Manufacturing Equipment	MFG	CNC Machines, Assembly Lines, Testing Equipment
Office Equipment	OFE	Printers, Scanners, Projectors, Phone Systems
Safety Equipment	SAF	Fire Extinguishers, First Aid, PPE, Emergency Lighting
Raw Materials	RAW	Metals, Plastics, Chemicals,

		Packaging Materials
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3.3 Role Permissions

Role	View Categories	Create Category	Edit Category	Delete Category
System Administrator	Yes	Yes	Yes	Yes
Warehouse Manager	Yes	Yes	Yes	No
Asset Custodian	Yes	No	No	No
Finance Officer	Yes	No	No	No
General User	Yes	No	No	No

4. Asset Lifecycle Management

Asset lifecycle management encompasses the complete journey of an asset from initial planning and acquisition through active use, maintenance, and eventual retirement or disposal. The NCG Warehouse Solutions platform provides comprehensive support for every stage of this lifecycle, ensuring full traceability and financial accountability throughout.

Understanding the asset lifecycle is critical for effective resource management. Each stage has distinct operational requirements, financial implications, and compliance considerations. The system enforces appropriate controls at each transition point, requiring proper authorisation, documentation, and audit trail entries.

The lifecycle stages supported by the system are: Planning (requirements identification), Acquisition (procurement and receipt), Commissioning (deployment and activation), Operation (active use and maintenance), Revaluation (periodic value assessment), Transfer (location or department changes), and Disposal (retirement, sale, or write-off).

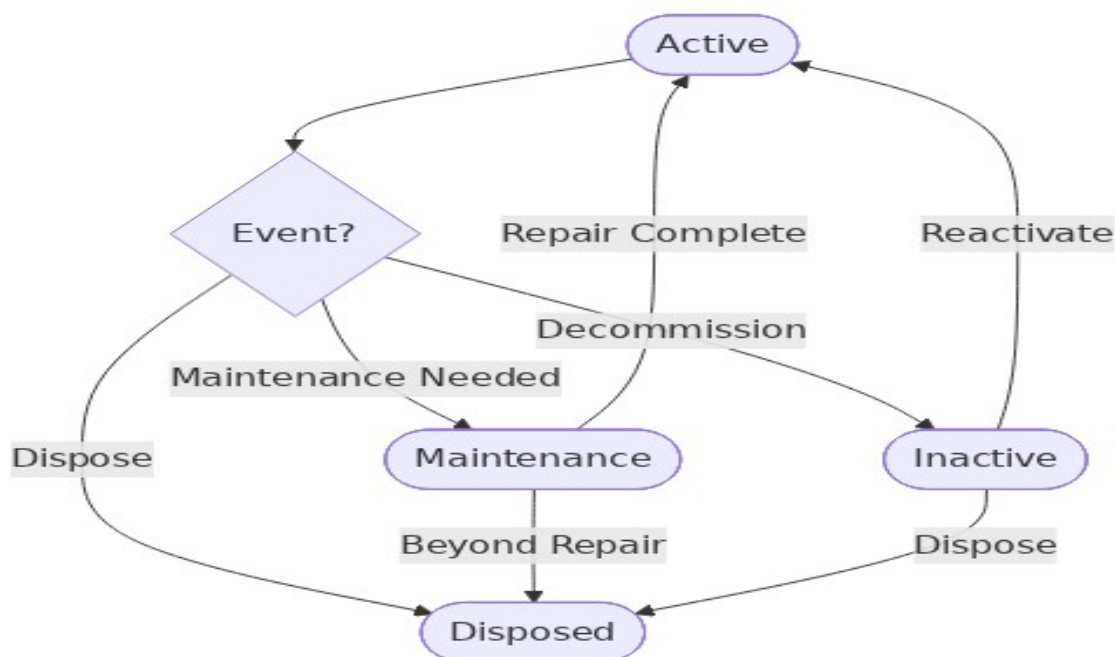


Figure 3: Complete Asset Lifecycle

4.1 Planning Stage

The planning stage begins when a department identifies the need for a new asset. This is typically initiated through the Asset Request system (covered in detail in Chapter 5). During planning, the requester must justify the need, estimate costs, identify the preferred asset type from the Asset Master catalogue, and specify the required delivery timeline.

Key considerations during planning include budget availability, alignment with organisational strategy, existing asset utilisation rates (are current assets being fully utilised before procuring new ones?), and total cost of ownership including maintenance, insurance, and eventual disposal costs.

4.2 Acquisition Stage

Once an asset request is approved, the acquisition process begins. This may involve procurement through the purchase order system, direct purchase, lease arrangement, or internal transfer from another department. The system tracks the acquisition method, cost, vendor, and all associated documentation.

Upon physical receipt, the asset is registered in the warehouse assets table with a unique asset tag, serial number, location assignment, and link to the originating Asset Master record. The system automatically generates the asset code following the INV-{CAT}-XXX convention.

4.3 Commissioning Stage

Commissioning involves preparing the asset for active service. This may include installation, configuration, testing, safety certification, and user training. The system records the commissioning date, any installation costs (which are capitalised as part of the asset cost per IAS 16), and the assigned custodian.

4.4 Operation Stage

During the operation stage, the asset is actively used for its intended purpose. The system tracks location, custodian, condition, maintenance history, and depreciation. Regular condition assessments should be recorded to support maintenance planning and impairment testing.

The system supports both scheduled (preventive) and unscheduled (corrective) maintenance tracking. Maintenance costs below the capitalisation threshold are expensed immediately, while significant improvements that extend the useful life or enhance capacity are capitalised per IAS 16.

4.5 Revaluation

Assets may be revalued periodically to reflect changes in fair market value. The revaluation module (Chapter 9) provides a structured process for recording value adjustments, calculating gains or losses, and generating the appropriate journal entries. Revaluation is particularly important for property, vehicles, and specialised equipment.

4.6 Transfer

Assets frequently move between locations, departments, or custodians during their operational life. The transfer module (Chapter 8) provides a controlled process for recording these movements with full audit trail. Transfers may trigger changes in cost centre allocation and require appropriate authorisations.

4.7 Disposal

When an asset reaches the end of its useful life, becomes obsolete, or is no longer needed, it enters the disposal process (Chapter 10). Disposal methods include sale, donation, recycling, or write-off. The system calculates the gain or loss on disposal, generates the required journal entries, and archives the asset record.

4.8 Status Lifecycle

Status	Description	Next Possible States	Triggered By
Planned	Asset requirement identified	Requested, Cancelled	Asset request creation
Requested	Formal request submitted	Approved, Rejected	Request submission
Approved	Request approved for procurement	Ordered, Cancelled	Approval workflow
Ordered	Purchase order issued	Received	PO creation
Received	Physically received at warehouse	Active, Under Inspection	GRN processing
Active	In operational use	Under Maintenance, Transferred, Disposed	Commissioning
Under Maintenance	Temporarily out of service	Active, Disposed	Maintenance request
Transferred	Moving between locations	Active	Transfer completion
Disposed	Permanently removed from service	Terminal state	Disposal approval

5. Asset Request and Approval Workflow

The Asset Request system provides a structured process for departments to request new assets, replacement assets, or additional quantities of existing asset types. Every request passes through a multi-stage approval workflow that ensures proper authorisation, budget verification, and procurement planning before any expenditure is committed.

The workflow is designed to balance operational agility with financial control. Standard requests follow a three-stage approval process: Department Head (HOD) approval, Procurement review, and final fulfilment. High-value requests above configurable thresholds may require additional approval stages such as Finance Director or CEO sign-off.

The system supports multiple request types within a single submission. A department can request a mix of new assets, replacements, and consumables in one request, each with its own justification and specifications. This reduces administrative overhead while maintaining individual tracking at the line-item level.

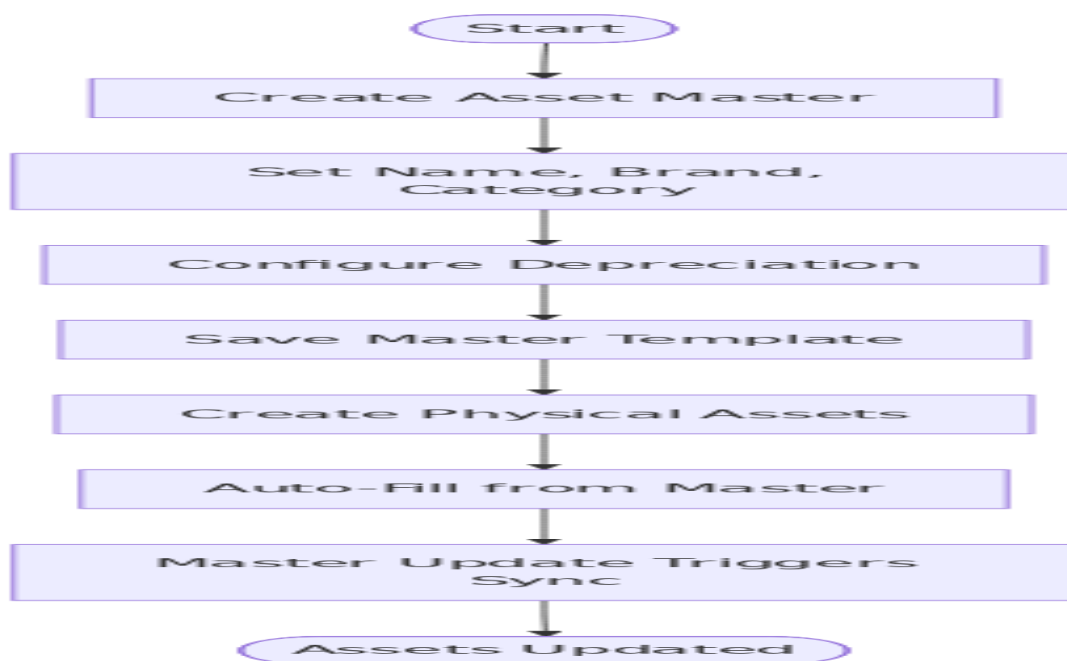


Figure 4: Asset Request Approval Workflow

5.1 What You Will See

The Asset Requests screen shows a list of all requests with columns for request number, requester name, department, priority, status, total estimated cost, and dates. Colour-coded status badges provide instant visual feedback: blue for pending, amber for in-review, green for approved, red for rejected.

Clicking on a request opens the detail view, which shows the full request information, line items, approval history, delivery tracking, and any attached documents. Action buttons change dynamically based on the current status and the viewer's role.

5.2 Creating a Request

5.2.1 Step-by-Step

- Navigate to Warehouse > Asset Management > Asset Requests
- Click 'New Request' to open the creation form
- Enter the requester details: name, department, contact number
- Specify the purpose and justification for the request
- Set the priority level: Low, Medium, High, or Urgent
- Set the required delivery date
- Add line items: select asset type from master catalogue or enter custom item details
- For each item, specify quantity, estimated unit price, and any special specifications
- Review the total estimated cost summary
- Click 'Submit Request' to enter the approval workflow

5.3 Approval Stages

Stage	Approver Role	Actions Available	Escalation
Stage 1: HOD Review	Department Head	Approve, Reject, Request Info	Auto-escalate after 3 days
Stage 2: Procurement	Procurement Officer	Approve, Reject, Partial Approve	Auto-escalate after 5 days
Stage 3: Finance (High Value)	Finance Director	Approve, Reject	Auto-escalate after 3 days
Stage 4: Fulfilment	Warehouse Manager	Fulfil from Stock, Mark for Purchase	No escalation

5.4 Fulfilment Methods

Once a request is fully approved, the fulfilment stage determines how each line item will be satisfied. The system supports two primary fulfilment methods:

- **Stock Fulfilment:** Items are issued from existing warehouse inventory. The system checks available stock levels and creates a material issue transaction.
- **Purchase Fulfilment:** Items are not available in stock and must be procured. The system creates a purchase requisition linked to the original asset request.

5.5 Delivery and Receipt

The delivery module tracks the physical handover of approved assets to the requesting department. Each delivery is recorded with the delivery date, location, items delivered, and the person receiving them. The requester must confirm receipt, which triggers the final status update on the request.

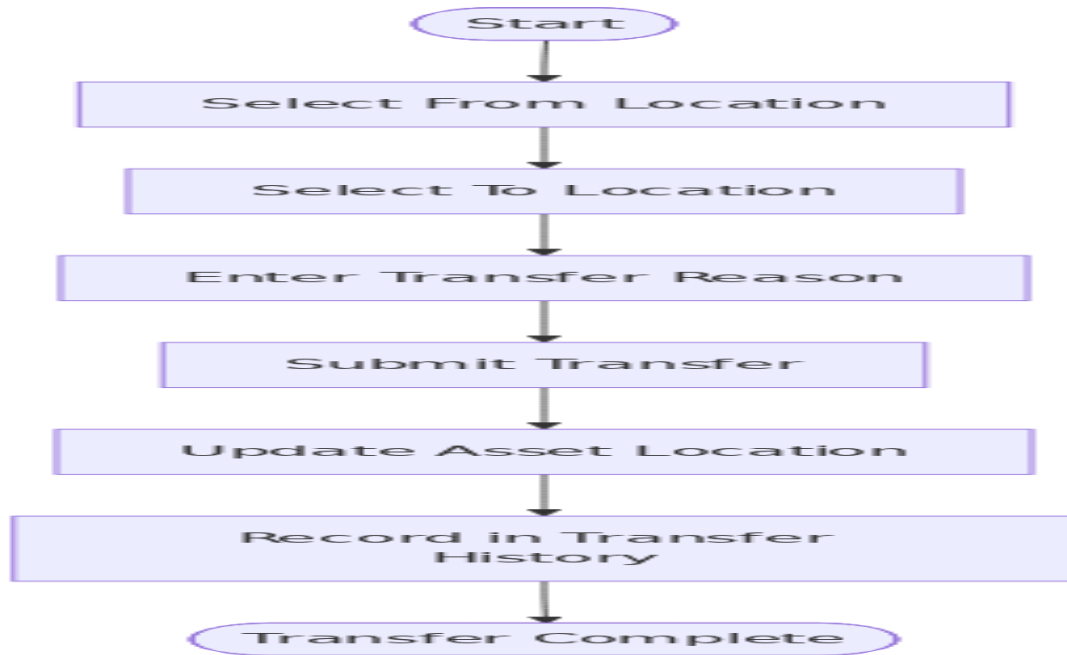


Figure 5: Asset Request Fulfilment Flow

5.6 Returns

If delivered assets are defective, incorrect, or no longer needed, the return process allows the requesting department to send items back to the warehouse. Returns are tracked against the original delivery and request, maintaining a complete chain of custody. Returned items are inspected and either restocked, sent for repair, or disposed of as appropriate.

6. Physical Asset Registry (Warehouse Assets)

The Warehouse Assets module is where individual physical assets are tracked throughout their operational life. While the Asset Master provides the template, each warehouse asset record represents a specific, individual asset with its own unique identifier, serial number, location, custodian, and condition history.

Every asset in the warehouse is assigned a unique asset tag using the INV-{CAT}-XXX code format. This code is printed as a QR code label and physically attached to the asset, enabling rapid identification and data access through mobile scanning. The QR code links directly to the asset's detail page, providing instant access to specifications, history, and maintenance records.

The warehouse asset record integrates data from multiple modules: procurement (how it was acquired), finance (depreciation and book value), maintenance (service history), and transfers (location history). This 360-degree view makes it the definitive system of record for each physical asset.

6.1 What You Will See

The Warehouse Assets screen provides a comprehensive list view with advanced filtering and sorting capabilities. Default columns include Asset Code, Asset Name, Category, Location, Department, Custodian, Status, Condition, Current Value, and Last Maintenance Date. Users can customise visible columns and save filter presets.

A card view option provides a visual layout showing the asset image, key details, and status badge. This view is particularly useful for visual identification during physical audits. The screen also includes bulk action capabilities for updating status, location, or custodian across multiple assets simultaneously.

6.2 Asset Registration

6.2.1 Automatic Code Generation

When creating a new warehouse asset, the system automatically generates the asset code based on the selected category. The format follows SAP MM material numbering standards: INV-{CategoryCode}-{Sequence}. For example, the third IT equipment asset registered would receive the code INV-ITE-003.

How Auto-Generation Works

1. User selects the asset category (e.g., IT Equipment with code ITE). 2. System queries existing assets with the INV-ITE- prefix within the current company. 3. Finds the highest existing sequence number (e.g., 042). 4. Generates the next code: INV-ITE-043. The code field is read-only during creation to prevent manual entry errors.

6.3 Location Management

Each asset is assigned to a specific location within the warehouse location hierarchy. Locations are structured as: Site > Building > Floor > Room > Rack/Shelf > Bin. This granular tracking enables rapid physical verification and efficient retrieval. Location changes are always recorded as transfers with full audit trail.

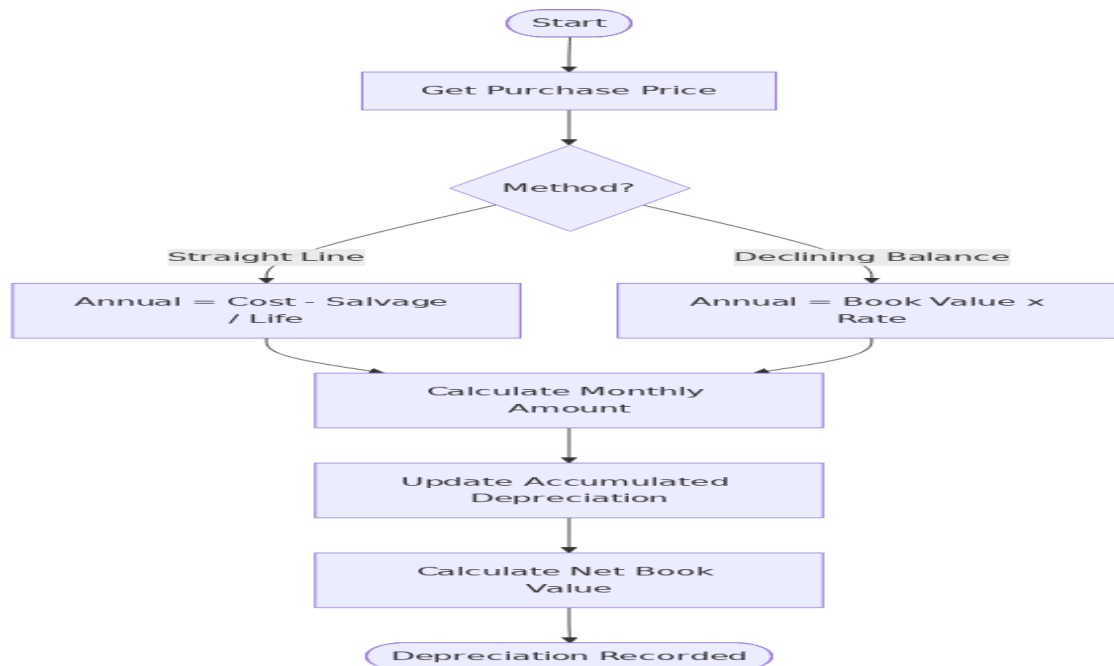


Figure 6: Asset Registration and Location Flow

6.4 Condition Assessment

Condition	Description	Depreciation Impact	Action Required
Excellent	New or like-new condition, fully functional	Standard rate	None
Good	Minor wear, fully functional	Standard rate	Routine maintenance
Fair	Moderate wear, functional with limitations	Consider accelerated	Scheduled maintenance
Poor	Significant wear, limited functionality	Accelerated rate	Major repair or disposal assessment
Non-Functional	Not operational, awaiting repair or disposal	Full impairment review	Immediate repair or disposal

7. Depreciation Engine

The depreciation engine is one of the most critical financial components of the Asset Management module. It automatically calculates and records the systematic allocation of an asset's cost over its useful life, in accordance with IAS 16 (Property, Plant and Equipment) and organisational accounting policies.

The system supports two primary depreciation methods: Straight-Line and Declining Balance. Each method is configured at the Asset Master level and inherited by individual warehouse assets, though individual assets can override the default method when warranted by specific circumstances.

Depreciation calculations run on a configurable schedule — typically monthly — and generate journal entries that are automatically posted to the general ledger. The process is fully auditable, with detailed logs showing the calculation inputs, method applied, and resulting figures for each asset.

7.1 Depreciation Methods

7.1.1 Straight-Line Method

The straight-line method allocates an equal amount of depreciation expense to each period over the asset's useful life. This is the most commonly used method and is appropriate for assets that provide consistent economic benefit throughout their life.

Formula

Annual Depreciation = (Cost - Salvage Value) / Useful Life Years
Monthly Depreciation = Annual Depreciation / 12

7.1.2 Worked Example: Straight-Line

A Dell Latitude 5540 laptop is purchased for \$1,200 with an estimated salvage value of \$200 and a useful life of 4 years.

Year	Opening Book Value	Depreciation	Closing Book Value
Year 1	\$1,200.00	\$250.00	\$950.00
Year 2	\$950.00	\$250.00	\$700.00
Year 3	\$700.00	\$250.00	\$450.00
Year 4	\$450.00	\$250.00	\$200.00

7.1.3 Declining Balance Method

The declining balance method applies a fixed depreciation rate to the remaining book value each period. This results in higher depreciation expense in the early years of the asset's life, which may better reflect the actual pattern of economic benefit consumption for certain asset types such as vehicles and technology equipment.

Formula

Annual Depreciation = Book Value at Start of Year x Depreciation Rate
The asset is never depreciated below its salvage value.

7.1.4 Worked Example: Declining Balance

A delivery vehicle is purchased for \$45,000 with a salvage value of \$8,000 and a depreciation rate of 30%.

Year	Opening Book Value	Depreciation (30%)	Closing Book Value
Year 1	\$45,000.00	\$13,500.00	\$31,500.00
Year 2	\$31,500.00	\$9,450.00	\$22,050.00
Year 3	\$22,050.00	\$6,615.00	\$15,435.00
Year 4	\$15,435.00	\$4,630.50	\$10,804.50
Year 5	\$10,804.50	\$2,804.50	\$8,000.00

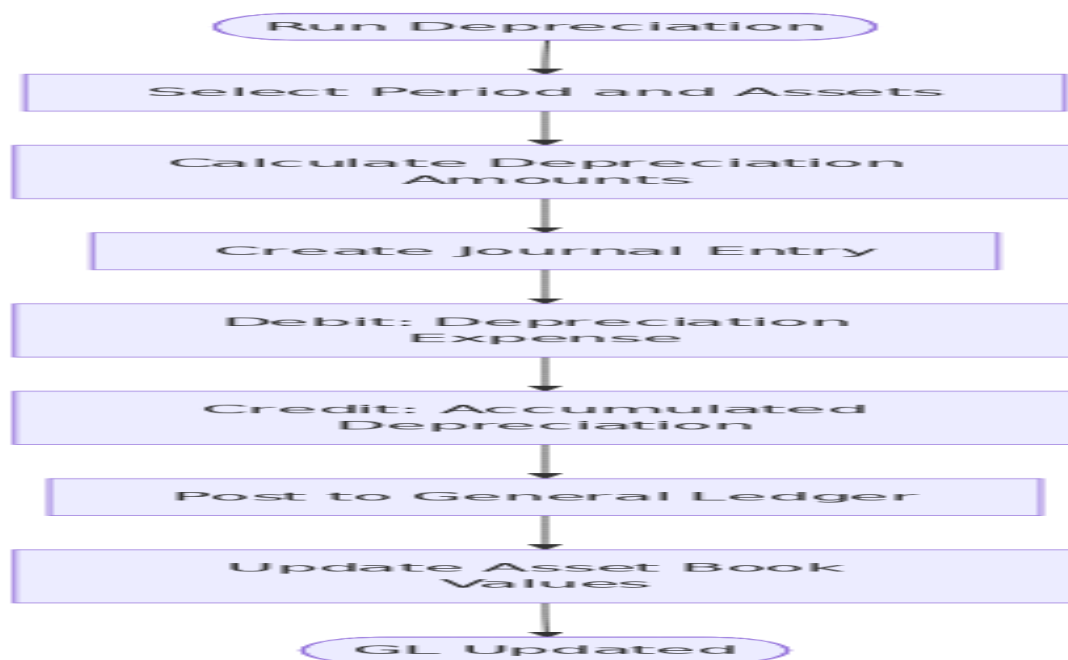


Figure 7: Depreciation Calculation Flow

7.2 Finance Integration

Each depreciation calculation generates a journal entry with two lines: a debit to the Depreciation Expense account (income statement) and a credit to the Accumulated Depreciation account (balance sheet contra-asset). These entries are automatically posted to the general ledger and linked back to the specific asset for full traceability.

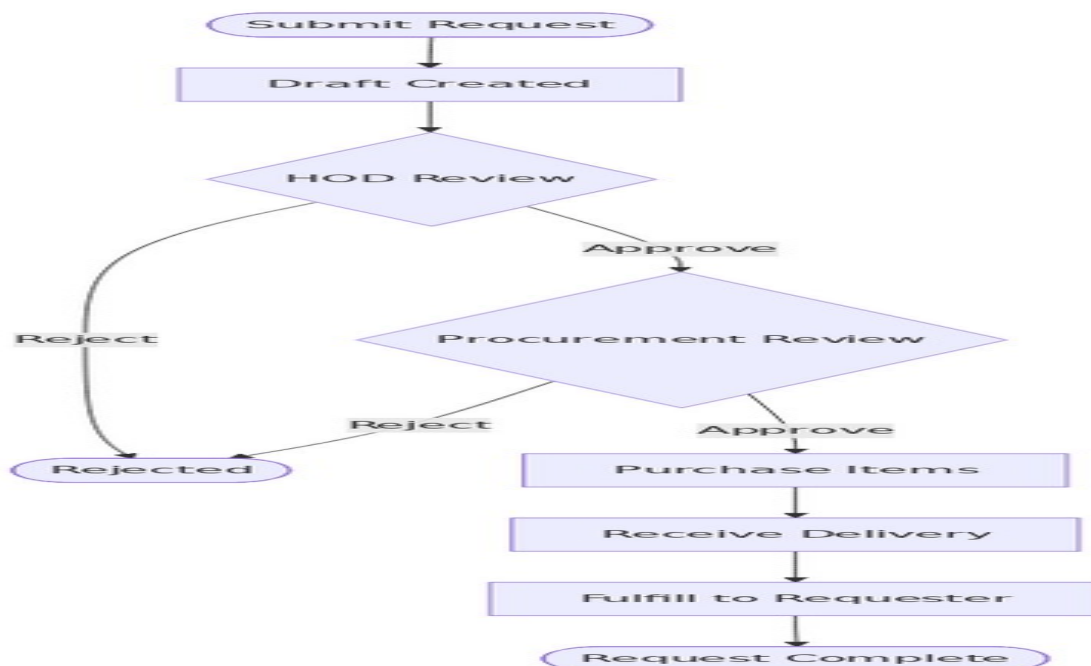


Figure 8: Depreciation to General Ledger Integration

7.3 Batch Processing

Monthly depreciation is processed as a batch operation that calculates depreciation for all active assets simultaneously. The batch process generates a summary report showing total depreciation by category, department, and cost centre, along with any exceptions or warnings (such as assets that have reached their salvage value or assets missing depreciation parameters).

7.4 Troubleshooting

Issue	Possible Cause	Resolution
Zero depreciation calculated	Asset has reached salvage value	Review if salvage value or useful life needs adjustment
Depreciation not running	Missing depreciation parameters	Ensure method, rate/useful life, and salvage value are set
Journal entry not posted	Accounting period is closed	Open the current accounting period or post to next period
Incorrect amount	Wrong method or rate configured	Review and correct depreciation parameters on asset master

8. Asset Transfers

Asset transfers record the movement of physical assets between locations, departments, or custodians. Every transfer creates an immutable audit trail entry that captures the before and after state, the reason for the transfer, and the person who authorised it. This is essential for maintaining accurate asset registers and satisfying audit requirements.

The transfer process supports multiple scenarios: physical relocation (moving a server from one data centre rack to another), departmental reassignment (transferring a vehicle from sales to operations), custodian changes (new employee taking responsibility for assigned equipment), and inter-company transfers within a group structure.

Each transfer captures the from and to values for location, sublocation, and department, along with the transfer date, reason, and any notes. The system validates that the destination location exists and has capacity before allowing the transfer to proceed.

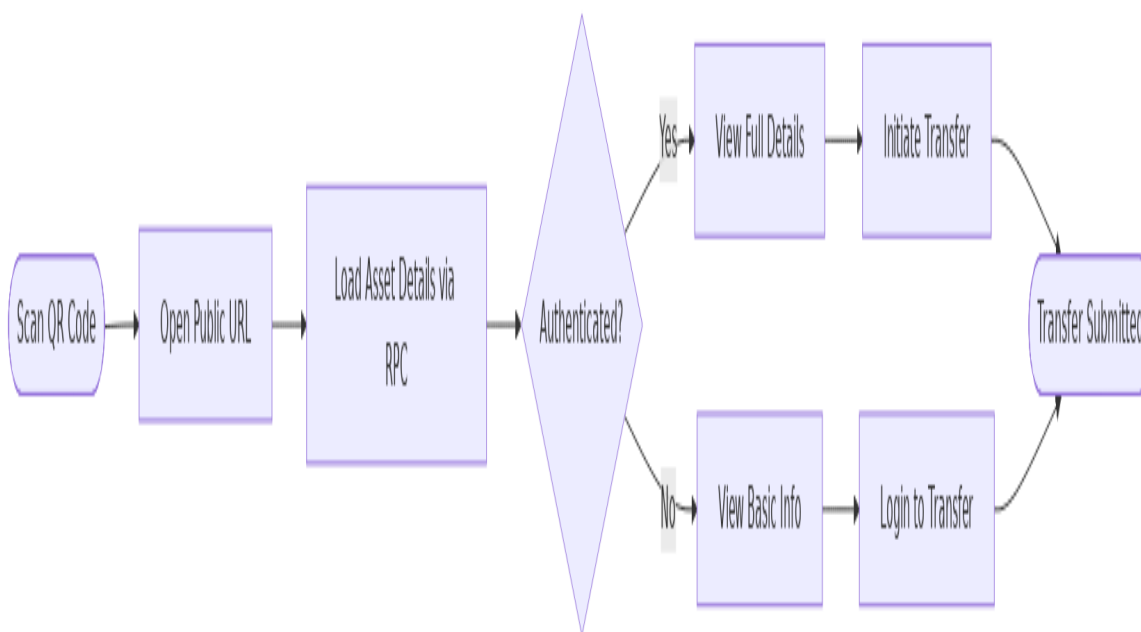


Figure 9: Asset Transfer Process Flow

8.1 What You Will See

The Transfer History screen shows a chronological list of all transfers with filterable columns for asset name, from location, to location, transfer date, reason, and transferred by. Each transfer record is read-only once created, ensuring the integrity of the audit trail.

8.2 Transfer Procedure

- Navigate to the asset detail page for the asset to be transferred
- Click the 'Transfer' action button
- Select the destination location from the location hierarchy picker
- Select the destination department (if changing)

- Enter the transfer reason and any notes
- Confirm the transfer date (defaults to today)
- Click 'Confirm Transfer' to execute

The system will update the asset's current location, department, and sublocation fields, and create a transfer history record. If the asset has a QR code label, it remains unchanged as the QR code links to the asset record (which now reflects the new location) rather than to the physical location itself.

8.3 Role Permissions

Role	Initiate Transfer	Approve Transfer	View History
System Administrator	Yes	Yes	Yes
Warehouse Manager	Yes	Yes	Yes
Asset Custodian	Yes (own assets)	No	Yes (own assets)
Department Head	Yes (dept assets)	Yes (dept assets)	Yes (dept assets)
Finance Officer	No	No	Yes

9. Asset Revaluation

Asset revaluation is the process of adjusting the carrying amount of an asset to reflect its current fair value. Under IAS 16, entities may choose either the cost model (carrying at cost less accumulated depreciation and impairment losses) or the revaluation model (carrying at fair value less subsequent accumulated depreciation and impairment losses).

The NCG Warehouse Solutions platform supports both models and provides a structured revaluation workflow. Revaluations must be performed with sufficient regularity that the carrying amount does not differ materially from fair value at the reporting date. For volatile asset classes such as property or specialised equipment, annual revaluations are recommended.

Each revaluation records the old carrying value, the new fair value, the resulting adjustment amount, the reason for revaluation, the date, and the person performing the revaluation. The system automatically generates the appropriate journal entries: revaluation surplus is credited to Other Comprehensive Income (revaluation reserve), while revaluation deficits are recognised in profit or loss unless they reverse a previous surplus.

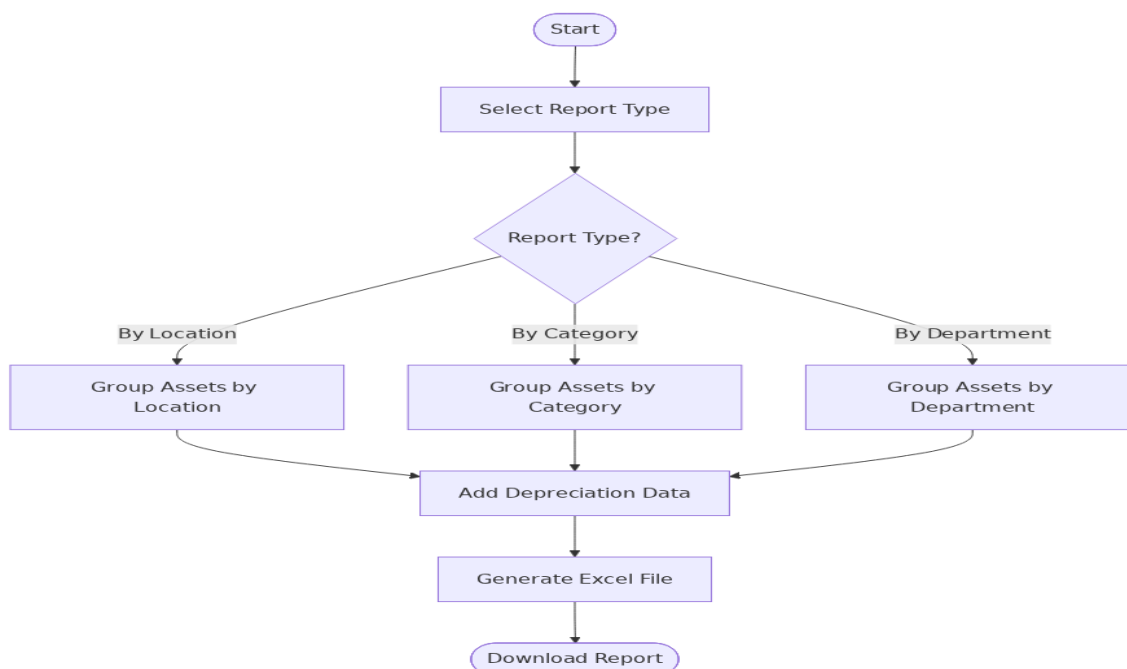


Figure 10: Asset Revaluation Workflow

9.1 Revaluation Procedure

- Select the asset to be revalued from the Warehouse Assets list
- Click the 'Revalue' action button
- Enter the new fair value based on professional valuation or market data
- Provide the reason for revaluation (e.g., market value increase, impairment)
- Review the calculated adjustment amount and accounting impact
- Confirm and submit for posting

9.2 Accounting Treatment

Scenario	Debit	Credit	Recognition
Revaluation increase (first time)	Asset Account	Revaluation Reserve (OCI)	Other Comprehensive Income
Revaluation increase (reversing prior decrease)	Asset Account	Profit & Loss	Income Statement (up to prior loss)
Revaluation decrease (first time)	Profit & Loss	Asset Account	Income Statement
Revaluation decrease (reversing prior increase)	Revaluation Reserve	Asset Account	Other Comprehensive Income (up to prior surplus)

10. Asset Disposal

Asset disposal is the process of removing an asset from active service and from the organisation's books. The disposal module handles the full range of disposal scenarios including sale to third parties, donation, recycling, trade-in against new equipment, and write-off for lost or destroyed assets.

The disposal process calculates the net book value at the date of disposal, compares it with any proceeds received, and determines the gain or loss on disposal. The resulting journal entry removes the asset cost and accumulated depreciation from the balance sheet and recognises the gain or loss in the income statement.

All disposals require appropriate approval based on the asset value and disposal method. The system enforces segregation of duties by requiring that the person approving the disposal is different from the person who initiated it. Environmental disposal requirements and data sanitisation procedures are tracked through the disposal notes.

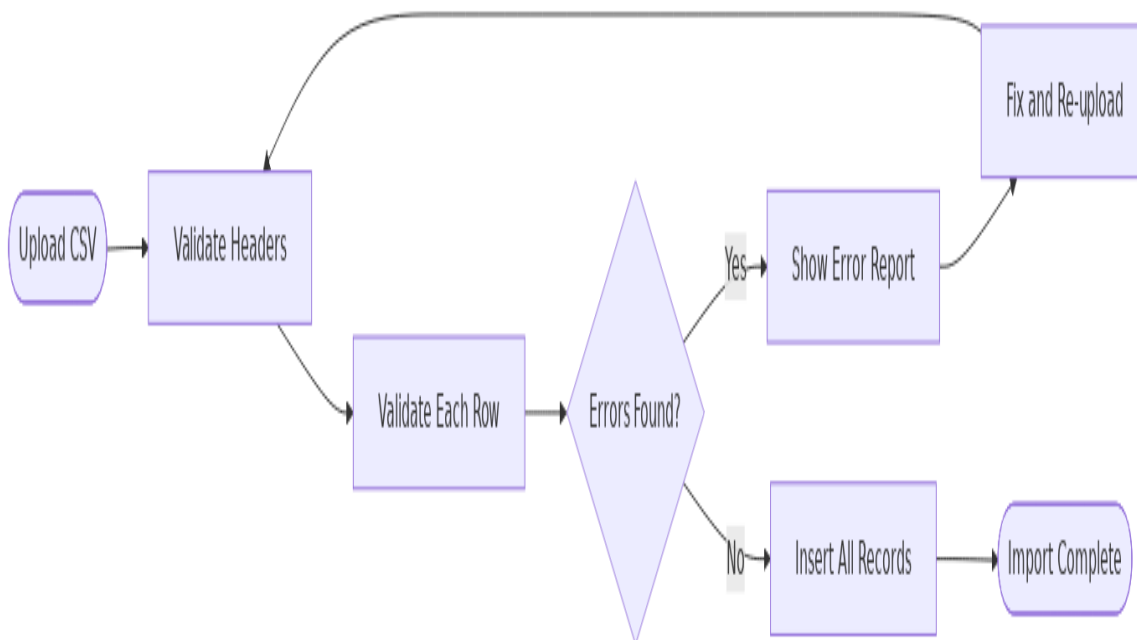


Figure 11: Asset Disposal Workflow

10.1 Disposal Methods

Method	Proceeds Expected	Documentation Required	Environmental Considerations
Sale	Yes - Market value	Sales agreement, buyer details, payment receipt	Transfer of environmental liability to buyer
Auction	Yes - Auction price	Auction listing, bid records, settlement	Buyer assumes environmental responsibility
Donation	No (or nominal)	Donation receipt,	Recipient accepts

		charity registration	environmental responsibility
Recycling	Possible scrap value	Recycling certificate, waste manifest	Certified recycling facility required
Write-Off	No	Write-off justification, management approval	Ensure proper disposal of hazardous materials
Trade-In	Credit against new purchase	Trade-in agreement, credit note	Dealer handles recycling/resale

10.2 Gain/Loss Calculation

Formula

Net Book Value = Original Cost - Accumulated Depreciation
 Gain/(Loss) on Disposal = Proceeds - Net Book Value
 Positive result = Gain (credited to income); Negative result = Loss (debited to expense)

10.3 Worked Example

A delivery vehicle was purchased for \$45,000 with accumulated depreciation of \$32,000. It is sold for \$15,000.

Item	Amount
Original Cost	\$45,000
Accumulated Depreciation	(\$32,000)
Net Book Value	\$13,000
Sale Proceeds	\$15,000
Gain on Disposal	\$2,000

11. QR Code Management

The QR Code Management feature provides a bridge between the digital asset registry and physical assets in the real world. Every warehouse asset can be assigned a QR code that, when scanned, provides instant access to the asset's complete record including specifications, location, custodian, maintenance history, and current book value.

QR codes are generated automatically when an asset is registered and can be printed as adhesive labels for physical attachment to the asset. The system supports both individual QR code generation and batch printing for large-scale asset tagging operations. Each QR code encodes a unique URL that resolves to the asset's detail page.

The QR scanning capability is particularly valuable during physical audits, maintenance inspections, and asset transfers. Field personnel can scan the code with any smartphone camera to immediately view and update the asset record, eliminating the need for paper-based forms or manual record lookup.

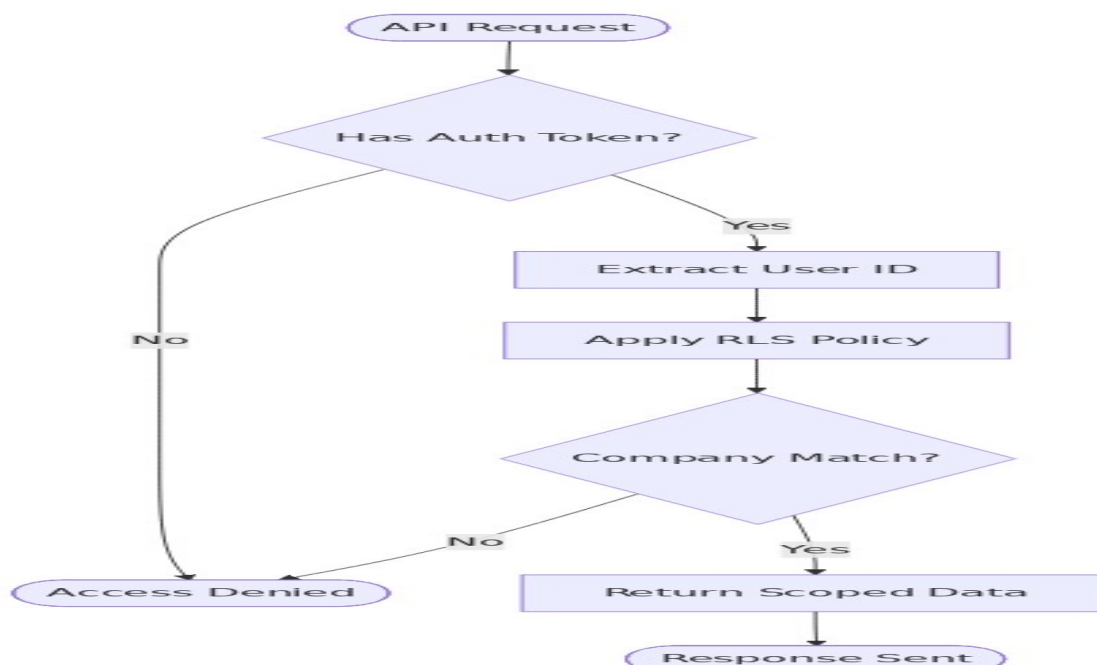


Figure 12: QR Code Scan and Access Flow

11.1 QR Code Generation

- QR codes are auto-generated using the qrcode library when an asset is created
- Each code encodes the URL: `https://{domain}/assets/scan/{asset_id}`
- The generated image is stored in Supabase Storage and linked to the asset record
- Batch generation supports printing labels for multiple assets simultaneously

11.2 Public Scan Experience

When a QR code is scanned by someone without system access, a simplified public view is displayed showing the asset name, code, category, location, and current custodian. This

facilitates asset identification without exposing confidential financial data. Authenticated users see the full detail view with all financial and operational information.

12. Reporting and Analytics

The reporting module provides comprehensive analytics and reporting capabilities for asset management. Reports are designed to serve multiple audiences: operational teams need asset lists and maintenance schedules, finance teams need depreciation summaries and book value reports, and management needs high-level dashboards with KPIs and trend analysis.

All reports support filtering by company, category, location, department, status, and date range. Reports can be exported in multiple formats including PDF, Excel (XLSX), and CSV for further analysis. Scheduled reports can be configured to run automatically and distribute via email.

12.1 Standard Reports

Report Name	Audience	Content	Frequency
Asset Register	All stakeholders	Complete list of all assets with key attributes	On demand
Depreciation Schedule	Finance	Monthly depreciation by asset, category, cost centre	Monthly
Book Value Summary	Finance / Management	Current book values by category and location	Quarterly
Asset Movement Report	Operations / Audit	All transfers within a specified period	Monthly
Disposal Report	Finance / Management	Disposed assets with gain/loss analysis	As needed
Maintenance Due Report	Operations	Assets requiring scheduled maintenance	Weekly
Audit Reconciliation	Audit / Finance	Physical vs. book comparison	Annual
Insurance Schedule	Risk Management	Assets requiring insurance coverage	Annual

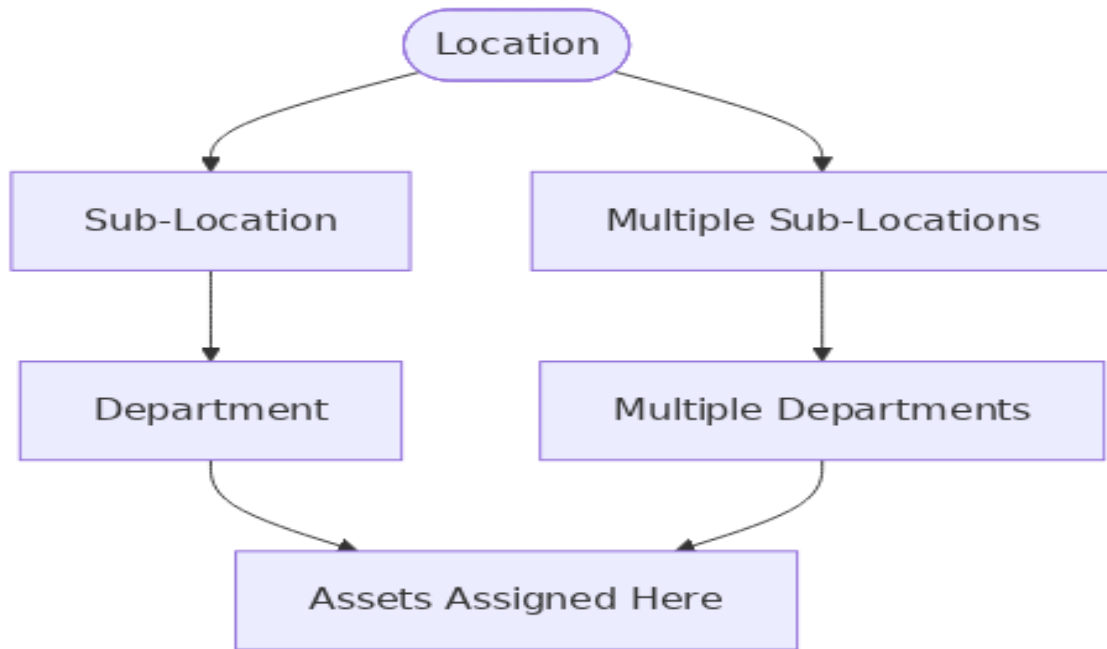


Figure 13: Reporting Pipeline

13. Security and Access Control

The Asset Management module implements a comprehensive security model based on Row-Level Security (RLS) in PostgreSQL. This ensures that users can only access data belonging to their assigned company, and can only perform actions permitted by their role. The security model follows the principle of least privilege, granting users the minimum access necessary to perform their duties.

Authentication is handled through Supabase Auth, supporting email/password login, magic links, and social login providers. All API requests are authenticated using JWT tokens, and database queries are filtered by `company_id` through RLS policies before any data is returned to the client.

Role-based access control (RBAC) is implemented through a dedicated `user_roles` table, separate from the user profile table, following security best practices to prevent privilege escalation attacks. Roles are checked server-side through security definer functions, never through client-side storage.

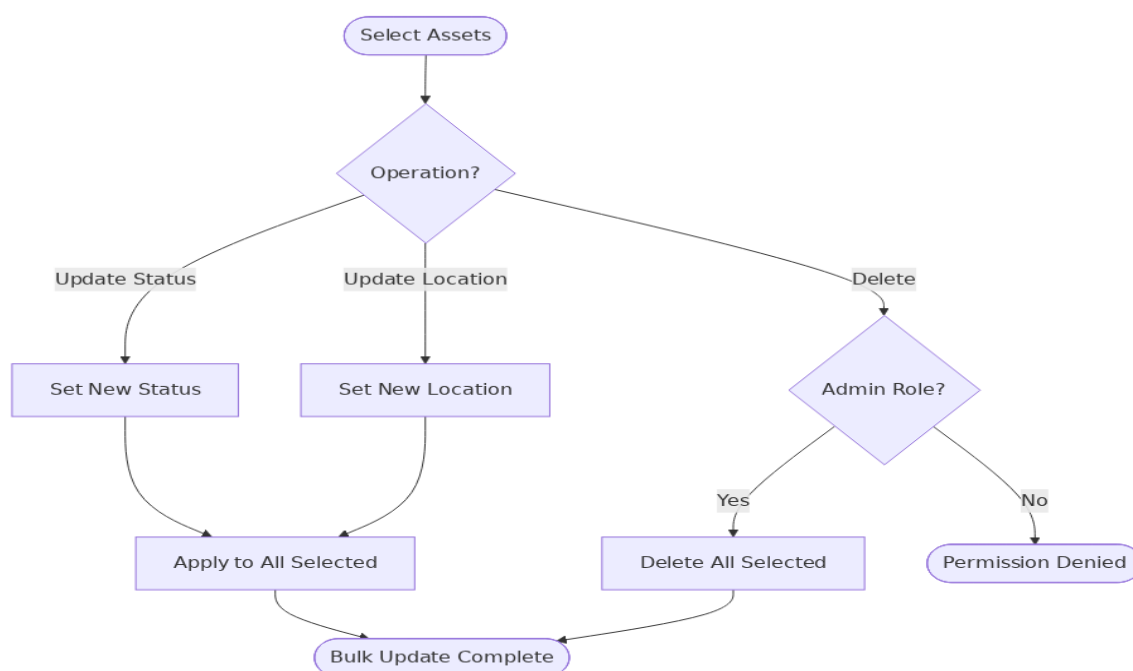


Figure 14: RLS Security Model

13.1 Role Definitions

Role	Description	Typical Users
System Administrator	Full access to all features and data within the company	IT Managers, System Owners
Warehouse Manager	Full access to asset operations, transfers, and disposals	Warehouse Supervisors, Asset Managers
Asset Custodian	View and update assigned assets, initiate transfers	Department Representatives, Equipment Custodians

Finance Officer	View all assets, manage depreciation, revaluation, and reporting	Accountants, Financial Controllers
Procurement Officer	View assets, manage asset requests and purchase orders	Buyers, Procurement Specialists
Auditor	Read-only access to all asset data and audit trails	Internal/External Auditors
General User	View own department assets, submit requests	All Employees

13.2 Data Isolation

Multi-tenant data isolation is enforced at the database level through RLS policies. Every asset-related table includes a `company_id` column, and RLS policies ensure that queries automatically filter by the authenticated user's company. This prevents any cross-company data leakage even if application-level bugs exist.

13.3 Audit Trail

Every significant action in the Asset Management module is logged in the audit trail. This includes asset creation, updates, transfers, revaluations, disposals, and approval workflow actions. Each audit entry records the action type, the user who performed it, the timestamp, the old and new values, and the IP address. Audit records are immutable and cannot be deleted or modified.

14. Maintenance Management

Effective maintenance management extends asset life, reduces total cost of ownership, and ensures operational continuity. The NCG Warehouse Solutions platform supports comprehensive maintenance tracking aligned with ISO 55001 asset management principles and industry best practices for preventive, corrective, and condition-based maintenance strategies.

The maintenance module integrates directly with the asset registry, allowing maintenance activities to be tracked against specific assets with full history. Maintenance costs are captured and compared against asset value thresholds to determine whether costs should be expensed (revenue expenditure) or capitalised (capital expenditure per IAS 16).

Preventive maintenance schedules are configured based on manufacturer recommendations, regulatory requirements, and organisational experience. The system generates work orders at configurable intervals (time-based or usage-based) and sends notifications to the responsible maintenance team or external service provider.

14.1 Maintenance Types

Type	Trigger	Planning	Cost Treatment
Preventive	Scheduled interval (time or usage)	Planned in advance	Usually expensed (routine)
Corrective	Equipment failure or defect	Unplanned / reactive	Expensed unless extends useful life
Condition-Based	Sensor data or inspection results	Predictive / planned	Expensed unless major overhaul
Capital Improvement	Enhancement or upgrade	Planned project	Capitalised (increases asset value)

14.2 Maintenance Workflow

- Work order is created (automatically from schedule or manually)
- Assigned to maintenance team or external vendor
- Parts and materials are requisitioned from warehouse stock
- Work is performed and documented
- Completion is recorded with actual costs, parts used, and findings
- Asset condition is reassessed and updated
- Costs are posted to the appropriate expense or capital account

14.3 Capitalisation Threshold

IAS 16 Guidance

Maintenance costs should be capitalised only when they: (a) extend the useful life beyond its original estimate, (b) increase the productive capacity, (c) improve the quality of output, or (d) reduce previously assessed operating costs. All other maintenance costs are expensed in the period incurred.

15. Insurance and Risk Management

Insurance tracking is a critical component of asset management that protects the organisation against financial loss from theft, damage, natural disasters, and other insurable events. The asset management module provides structured tracking of insurance policies linked to individual assets or asset groups.

Risk management for physical assets follows the ISO 31000 framework, identifying potential threats to asset availability, integrity, and value. The system supports risk assessment matrices that combine likelihood and impact to prioritise risk mitigation activities.

Each asset can have associated insurance details including policy number, insurer, coverage amount, premium, expiry date, and coverage type. The system generates alerts when policies are approaching renewal dates and flags assets that lack adequate insurance coverage.

15.1 Insurance Coverage Types

Coverage Type	Covers	Typical Assets
All Risks	Physical damage from any cause except exclusions	High-value equipment, vehicles
Fire and Theft	Fire damage and theft/burglary	Office equipment, inventory
Transit	Damage during transportation	Mobile assets, deliveries
Business Interruption	Loss of income due to asset failure	Critical production equipment
Liability	Third-party injury or property damage	Vehicles, public-facing assets
Cyber	Data loss or cyber attack on IT assets	Servers, network equipment

15.2 Risk Assessment Matrix

Likelihood / Impact	Negligible	Minor	Moderate	Major	Catastrophic
Almost Certain	Medium	High	High	Critical	Critical
Likely	Medium	Medium	High	High	Critical
Possible	Low	Medium	Medium	High	High
Unlikely	Low	Low	Medium	Medium	High
Rare	Low	Low	Low	Medium	Medium

16. Audit and Compliance

Regular physical audits ensure that the asset register accurately reflects the assets actually held by the organisation. Discrepancies between the physical count and the book record may indicate theft, loss, damage, misclassification, or administrative errors. The system provides structured support for planning, executing, and reconciling physical asset audits.

The audit process is designed to comply with ISA 501 (Audit Evidence regarding Specific Items) and organisational internal audit standards. Auditors can use the QR scanning capability to rapidly verify assets against the register, recording any discrepancies for investigation and resolution.

Compliance monitoring extends beyond physical audits to include ongoing verification of depreciation accuracy, insurance coverage adequacy, maintenance schedule adherence, and policy compliance across all asset management activities.

16.1 Physical Audit Procedure

- Generate the Asset Register report for the audit scope (location, category, or department)
- Print or download the audit checklist with asset codes and expected locations
- Visit each location and physically verify each asset using QR code scanning
- Record the actual condition, location, and any discrepancies
- For missing assets, initiate investigation workflow
- For found but unregistered assets, create new records or investigate source
- Generate the Audit Reconciliation Report comparing physical to book
- Adjust the register for confirmed discrepancies with proper authorisation
- File the audit report and management response for regulatory compliance

16.2 Compliance Checklist

Compliance Area	Requirement	Frequency	Evidence
Physical Verification	Count and verify all assets	Annual	Audit reconciliation report
Depreciation Accuracy	Verify calculations match policy	Quarterly	Depreciation schedule review
Insurance Coverage	Verify all assets adequately covered	Annual	Insurance schedule vs asset register
Disposal Authorisation	Verify all disposals properly approved	Per occurrence	Approval records, disposal forms
Access Control Review	Review user roles and permissions	Semi-annual	Role assignment report
Data Backup Verification	Verify asset data is backed up	Monthly	Backup logs, restoration tests

17. Integration Architecture

The Asset Management module does not operate in isolation. It integrates with multiple modules within the NCG Warehouse Solutions platform and can connect to external systems through well-defined interfaces. Understanding these integration points is essential for system administrators and developers responsible for maintaining data flow integrity.

Internal integrations include the Procurement module (purchase orders and goods receipts feed asset creation), the Finance module (depreciation, revaluation, and disposal generate journal entries), the Inventory module (consumable assets are managed through stock transactions), and the HR module (employee records link to asset custodianship).

External integration is supported through the Supabase REST API and edge functions. Third-party systems such as ERP platforms, CMMS (Computerised Maintenance Management Systems), and insurance management systems can read and write asset data through authenticated API endpoints.

17.1 Internal Integration Map

Source Module	Integration Point	Data Flow	Trigger
Procurement	Purchase Order Completion	PO items create warehouse assets	GRN confirmation
Finance	Depreciation Posting	Depreciation entries to GL	Monthly batch process
Finance	Disposal Posting	Gain/loss entries to GL	Disposal approval
Finance	Revaluation Posting	Revaluation entries to GL	Revaluation confirmation
Inventory	Stock Issue/Receipt	Consumable asset movements	Material issue/receipt
HR	Employee Assignment	Custodian link to employee record	Asset assignment change
Maintenance	Work Order	Maintenance costs to asset history	Work order completion

17.2 API Endpoints

Endpoint	Method	Description	Auth Required
/rest/v1/warehouse_assets	GET	List all assets (filtered by RLS)	JWT Bearer Token
/rest/v1/warehouse_assets	POST	Create new asset record	JWT Bearer Token
/rest/v1/asset_transfers	POST	Record asset transfer	JWT Bearer Token
/rest/v1/asset_disposals	POST	Record asset disposal	JWT Bearer Token
/rest/v1/	POST	Record asset	JWT Bearer Token

asset_revaluations		revaluation	
/rest/v1/ asset_transactions	GET	Query transaction history	JWT Bearer Token
/functions/v1/generate- qr	POST	Generate QR code for asset	JWT Bearer Token
/functions/v1/calculate- depreciation	POST	Run depreciation calculation	JWT Bearer Token

18. CI/CD Pipeline and Deployment

The Asset Management module is deployed through a structured CI/CD pipeline that ensures code quality, automated testing, and controlled releases. The pipeline is integrated with the Lovable development platform, which provides automatic builds, preview deployments, and production publishing.

Database schema changes are managed through Supabase migrations, which provide version-controlled, reversible database evolution. Each migration is tested in a preview environment before being applied to production, ensuring that schema changes do not break existing functionality.

Edge functions are deployed independently from the frontend application, allowing server-side logic to be updated without requiring a full application redeployment. This separation of concerns enables rapid iteration on business logic while maintaining frontend stability.

18.1 Deployment Pipeline Stages

Stage	Environment	Purpose	Gate
Development	Local / Lovable Preview	Feature development and initial testing	Code review
Preview	Preview URL	Integration testing and stakeholder review	QA approval
Staging	Staging environment	Performance testing and UAT	UAT sign-off
Production	Published URL	Live production deployment	Release manager approval

18.2 Database Migration Strategy

All database changes follow a migration-first approach. New tables, columns, indexes, and RLS policies are defined as SQL migration files in the supabase/migrations directory. Each migration is timestamped and applied in order. Rollback scripts are maintained for critical changes to enable rapid recovery if issues are discovered post-deployment.

18.3 Edge Function Architecture

Edge functions run on Deno Deploy (via Supabase) and handle server-side operations that require elevated privileges, complex calculations, or external service integration. Key asset management edge functions include QR code generation, batch depreciation processing, notification dispatch, and report generation.

Function	Trigger	Purpose	Runtime
generate-qr	Asset creation / manual trigger	Generate QR code PNG and store in bucket	< 2 seconds
calculate-depreciation	Monthly scheduler / manual	Calculate depreciation for all active assets	< 30 seconds

notify-maintenance	Maintenance schedule trigger	Send maintenance due notifications	< 5 seconds
export-asset-report	User request	Generate PDF/Excel asset reports	< 15 seconds

Appendix A: Glossary of Terms

Term	Definition
Accumulated Depreciation	Total depreciation charged against an asset since acquisition
Asset Custodian	Person responsible for the care and safekeeping of an assigned asset
Asset Master	Template record defining the standard attributes of an asset type
Book Value (NBV)	Original cost minus accumulated depreciation and impairment losses
Capitalisation	Recording expenditure as an asset rather than an expense
Capitalisation Threshold	Minimum value above which expenditure must be capitalised as an asset
Depreciation	Systematic allocation of an asset's cost over its useful life
Disposal	Process of removing an asset from the register (sale, write-off, donation)
Fair Value	Price receivable in an orderly transaction between market participants
GRN	Goods Received Note — document confirming physical receipt of purchased items
IAS 16	International Accounting Standard for Property, Plant and Equipment
Impairment	Reduction in recoverable amount below carrying amount
ISO 55001	International standard for Asset Management Systems
RLS	Row-Level Security — database-level access control mechanism
Salvage Value	Estimated residual value at end of useful life
Straight-Line	Depreciation method allocating equal expense each period
Useful Life	Period over which an asset is expected to provide economic benefit
Warehouse Asset	Individual physical asset tracked in the warehouse system

Appendix B: Database Schema Reference

This appendix provides the complete database schema for all asset-related tables in the system.

B.1 asset_master

Column	Type	Nullable	Description
id	UUID	No	Primary key
asset_name	TEXT	No	Name of the asset type
brand	TEXT	Yes	Manufacturer or brand
category_id	UUID FK	Yes	Reference to asset_categories
subcategory_id	UUID FK	Yes	Reference to asset_categories (subcategory)
purchase_price	NUMERIC	Yes	Standard purchase price
current_value	NUMERIC	Yes	Current estimated value
depreciation_method	TEXT	Yes	Straight-Line or Declining Balance
depreciation_rate	NUMERIC	Yes	Annual depreciation rate (%)
useful_life_years	NUMERIC	Yes	Expected useful life in years
salvage_value	NUMERIC	Yes	Residual value at end of life
purchase_date	DATE	Yes	Date of purchase
accumulated_depreciation	NUMERIC	Yes	Total depreciation to date
image_url	TEXT	Yes	URL to asset image in storage
description	TEXT	Yes	Free-text description
status	TEXT	Yes	Active or Inactive
company_id	UUID FK	Yes	Company (multi-tenant)
created_by	UUID FK	Yes	User who created the record
created_at	TIMESTAMPTZ	No	Creation timestamp
updated_at	TIMESTAMPTZ	No	Last update timestamp

B.2 warehouse_assets

Column	Type	Nullable	Description
id	UUID	No	Primary key
asset_name	TEXT	No	Name/description of the physical asset
asset_tag	TEXT	Yes	Physical tag/label identifier
serial_number	TEXT	Yes	Manufacturer serial number
asset_master_id	UUID FK	Yes	Reference to asset_master template
category_id	UUID FK	Yes	Asset category
location_id	UUID FK	Yes	Current warehouse location
department_id	UUID FK	Yes	Assigned department
status	TEXT	Yes	Current lifecycle status
condition	TEXT	Yes	Physical condition assessment
purchase_price	NUMERIC	Yes	Actual purchase price for this unit
current_value	NUMERIC	Yes	Current book value
company_id	UUID FK	Yes	Company (multi-tenant)
qr_code_url	TEXT	Yes	URL to generated QR code image

B.3 asset_disposals

Column	Type	Nullable	Description
id	UUID	No	Primary key
asset_id	UUID FK	No	Reference to warehouse_assets
disposal_date	DATE	No	Date of disposal
disposal_method	TEXT	No	Sale, Donation, Write-Off, etc.
proceeds	NUMERIC	No	Amount received from disposal
net_book_value_at_disposal	NUMERIC	No	NBV at disposal date

gain_loss	NUMERIC	Yes	Calculated gain or loss
buyer_name	TEXT	Yes	Name of buyer (if sold)
notes	TEXT	Yes	Additional disposal notes
status	TEXT	No	Pending, Approved, Completed
journal_entry_id	UUID FK	Yes	Link to finance journal entry

B.4 asset_transfers

Column	Type	Nullable	Description
id	UUID	No	Primary key
asset_id	UUID FK	No	Reference to warehouse_assets
from_location_id	UUID FK	Yes	Origin location
to_location_id	UUID FK	Yes	Destination location
from_department_id	UUID FK	Yes	Origin department
to_department_id	UUID FK	Yes	Destination department
transfer_date	DATE	No	Date of transfer
transfer_reason	TEXT	Yes	Reason for transfer
transferred_by	UUID FK	Yes	User who performed transfer

B.5 asset_revaluations

Column	Type	Nullable	Description
id	UUID	No	Primary key
asset_id	UUID FK	No	Reference to warehouse_assets
old_value	NUMERIC	No	Carrying value before revaluation
new_value	NUMERIC	No	Fair value after revaluation
adjustment_amount	NUMERIC	Yes	Difference (new - old)
reason	TEXT	Yes	Reason for revaluation
revaluation_date	DATE	No	Date of revaluation
revalued_by	UUID FK	Yes	User who performed

			revaluation
journal_entry_id	UUID FK	Yes	Link to finance journal entry

Appendix C: ISO 55001 Compliance Checklist

ISO 55001 Clause	Requirement	System Feature	Status
4.1	Understanding the organisation context	Company setup, multi-tenant configuration	Implemented
4.2	Needs and expectations of stakeholders	Role-based dashboards and reports	Implemented
5.1	Leadership and commitment	Approval workflows, executive dashboards	Implemented
6.1	Actions to address risks and opportunities	Risk assessment matrix, insurance tracking	Implemented
6.2	Asset management objectives	KPI dashboards, depreciation targets	Implemented
7.1	Resources	Budget tracking, procurement integration	Implemented
7.5	Information requirements	Comprehensive audit trails, document management	Implemented
8.1	Operational planning and control	Maintenance scheduling, work orders	Implemented
8.2	Management of change	Transfer workflows, revaluation process	Implemented
9.1	Monitoring, measurement, analysis	Analytics dashboards, automated reporting	Implemented
9.2	Internal audit	Physical audit module, reconciliation reports	Implemented
10.1	Nonconformity and corrective action	Issue tracking, discrepancy resolution workflows	Implemented
10.2	Preventive action	Condition-based maintenance, predictive analytics	Implemented
10.3	Continual improvement	Version-controlled processes, feedback mechanisms	Implemented

Appendix D: Standard Operating Procedures

D.1 SOP-AM-001: New Asset Registration

Field	Value
Procedure ID	SOP-AM-001
Version	2.0
Effective Date	April 2026
Review Date	April 2027
Owner	Warehouse Manager
Approved By	Operations Director

Purpose: To ensure all new assets are registered accurately and consistently in the NCG Warehouse Solutions system, with complete financial and operational data captured at the point of entry.

Scope: This procedure applies to all physical assets received at any warehouse or office location within the organisation, regardless of acquisition method (purchase, donation, transfer, or lease).

Procedure Steps

1. Verify physical receipt of asset against delivery note / purchase order
2. Inspect asset for damage and record condition assessment
3. Verify asset specifications match the purchase order or request
4. Navigate to Warehouse > Assets > Create New Asset
5. Select the Asset Master template (or create new if first instance)
6. Enter serial number, model number, and any unique identifiers
7. Assign location using the location hierarchy picker
8. Assign custodian (department head or designated individual)
9. Enter actual purchase price and purchase date
10. Generate and attach QR code label to physical asset
11. Photograph the asset and upload to the system
12. Save the record and verify the auto-generated asset code

D.2 SOP-AM-002: Periodic Depreciation Processing

Field	Value
Procedure ID	SOP-AM-002
Version	2.0
Effective Date	April 2026
Review Date	April 2027

Owner	Finance Manager
Approved By	CFO

Purpose: To ensure monthly depreciation is calculated and posted accurately for all active assets.

Procedure Steps

- 1. Verify that all new assets for the period have been registered with correct depreciation parameters
- 2. Verify the current accounting period is open
- 3. Navigate to Finance > Depreciation > Run Batch
- 4. Select the accounting period for depreciation processing
- 5. Review the preview showing assets to be depreciated and calculated amounts
- 6. Investigate and resolve any exceptions or warnings
- 7. Execute the batch process
- 8. Review the generated journal entries for accuracy
- 9. Post the journal entries to the general ledger
- 10. Generate the Depreciation Schedule report for the period
- 11. File the report and obtain supervisor sign-off

D.3 SOP-AM-003: Annual Physical Audit

Field	Value
Procedure ID	SOP-AM-003
Version	2.0
Effective Date	April 2026
Review Date	April 2027
Owner	Internal Audit Manager
Approved By	CEO

Purpose: To verify the existence, condition, and location of all physical assets against the asset register.

Procedure Steps

- 1. Define audit scope: locations, categories, and asset value thresholds
- 2. Notify department heads of upcoming audit timeline
- 3. Generate Asset Register report for audit scope
- 4. Assign audit teams to locations with printed checklists
- 5. Physically visit each location and scan asset QR codes
- 6. Record actual condition, location, and any discrepancies
- 7. Investigate missing assets: check transfer records, query custodians
- 8. Investigate unregistered assets: verify source and create records if valid

- 9. Generate Reconciliation Report with findings
- 10. Present findings to management for resolution decisions
- 11. Implement approved adjustments with proper documentation
- 12. File audit report and management response

Appendix E: Troubleshooting Guide

E.1 Common Issues and Resolutions

Issue	Symptoms	Cause	Resolution
Asset code not generated	Code field shows 'Select category'	No category selected	Select a Level 1 category first
Cannot create asset	Save button disabled	Required fields missing	Complete all mandatory fields (name, category)
Depreciation shows zero	Monthly batch shows \$0	Asset reached salvage value	Review useful life/salvage value settings
Transfer fails	Error: Invalid destination	Location does not exist	Create the destination in Location Management
QR code not scanning	Camera does not recognise code	Low print quality	Regenerate QR code and reprint at higher DPI
Report shows no data	Empty report generated	Filter too restrictive	Broaden date range or remove category filter
Cannot approve request	Approve button not visible	Insufficient permissions	Contact admin to assign the approver role
Journal entry not posted	Entry stuck in draft	Period is closed	Open the period or post to next open period
Duplicate asset records	Same serial number twice	Manual entry error	Merge records and enable duplicate detection
Slow report generation	Report takes > 30 seconds	Large dataset without filters	Apply date/category filters to reduce scope

E.2 Error Codes

Code	Message	Action
AM-001	Asset not found	Verify asset ID or code; check company context
AM-002	Insufficient permissions	Contact system administrator for role assignment
AM-003	Invalid depreciation parameters	Ensure method, rate, useful life, and salvage are set
AM-004	Transfer destination invalid	Verify location exists and is

		active
AM-005	Disposal approval required	Submit for approval before processing
AM-006	Accounting period closed	Open the period or select next open period
AM-007	Duplicate serial number	Check existing records for the same serial number
AM-008	QR generation failed	Retry; check edge function logs if persistent

Appendix F: System Configuration Guide

F.1 Initial System Setup

Before the Asset Management module can be used effectively, several configuration steps must be completed by the system administrator. These settings establish the foundation for all asset tracking, financial calculations, and reporting functions. Proper configuration is critical for ensuring data accuracy and compliance with organisational policies.

The configuration process should be completed in the order presented below, as some settings depend on others. For example, asset categories must be configured before Asset Master records can be created, and GL account mappings must be established before depreciation posting can function correctly.

F.1.1 Company Configuration

Each company within the multi-tenant system requires its own configuration. Navigate to Administration > Company Settings to configure the following parameters for each company:

Company Name and Registration: Enter the legal entity name, registration number, tax identification number, and registered address. These details appear on generated documents and reports.

Financial Year Settings: Define the fiscal year start month and the accounting period structure (typically 12 monthly periods). This drives the depreciation schedule and period-end reporting.

Currency Configuration: Set the base currency for the company. All asset values, depreciation calculations, and financial reports use this currency. Multi-currency support is available for organisations operating across borders.

Approval Thresholds: Configure the value thresholds that determine which approval stages are required for asset requests, disposals, and revaluations. For example, requests above \$10,000 may require Finance Director approval in addition to HOD and Procurement stages.

F.1.2 Location Hierarchy Setup

The location hierarchy defines the physical structure of the organisation's facilities. A well-designed location hierarchy enables precise asset tracking and efficient physical audits. The recommended hierarchy follows this structure:

Level 1 - Site: The highest level, representing a geographical location or campus. Examples: Head Office, Factory Site A, Distribution Centre North.

Level 2 - Building: Individual buildings within a site. Examples: Main Office Building, Warehouse Block A, Production Hall.

Level 3 - Floor/Area: Floors within a building or designated areas. Examples: Ground Floor, Mezzanine Level, Loading Bay Area.

Level 4 - Room/Zone: Specific rooms or operational zones. Examples: Server Room, Conference Room B, Assembly Line 3.

Level 5 - Rack/Shelf/Bay: Detailed storage positions for small or warehouse-stored assets. Examples: Rack A-14, Shelf 3B, Bay 7.

When designing the location hierarchy, consider future expansion. It is much easier to add new locations within an existing structure than to reorganise the hierarchy after assets have been assigned. Each location should have a unique, meaningful code that staff can easily remember and reference.

F.1.3 Category Hierarchy Configuration

Asset categories must be configured before any assets can be registered. The category hierarchy typically mirrors the organisation's chart of accounts structure for fixed assets, ensuring alignment between operational tracking and financial reporting.

Each Level 1 category requires a three-letter mnemonic code that forms part of the auto-generated asset code (INV-{CODE}-XXX). Choose codes that are intuitive and memorable. Once assigned and assets have been created with that code, the category code should not be changed to maintain referential integrity.

Subcategories provide operational granularity without affecting the coding structure. For example, under 'IT Equipment' (ITE), you might have subcategories for Laptops, Desktops, Servers, Networking Equipment, and Peripherals. This enables detailed reporting while keeping the asset code structure simple.

F.1.4 GL Account Mapping

The General Ledger account mapping connects asset management transactions to the correct financial accounts. The following mappings must be configured for each asset category:

Asset Account: The balance sheet account where the asset's cost is recorded (e.g., 1510 - Property, Plant & Equipment).

Accumulated Depreciation Account: The contra-asset account for accumulated depreciation (e.g., 1519 - Accumulated Depreciation).

Depreciation Expense Account: The income statement account for depreciation charges (e.g., 6100 - Depreciation Expense).

Gain on Disposal Account: Income statement account for gains on asset disposal (e.g., 7200 - Gain on Disposal of Assets).

Loss on Disposal Account: Income statement account for losses on disposal (e.g., 6200 - Loss on Disposal of Assets).

Revaluation Reserve Account: Equity account for revaluation surpluses (e.g., 3200 - Revaluation Reserve).

These mappings ensure that all automated journal entries generated by the depreciation engine, disposal process, and revaluation module post to the correct accounts without manual intervention.

F.2 User and Role Configuration

User access to the Asset Management module is controlled through role assignments. The system administrator must assign appropriate roles to each user based on their job responsibilities. Roles can be combined — for example, a warehouse supervisor might be assigned both the Warehouse Manager and Asset Custodian roles.

When assigning roles, follow the principle of least privilege: grant only the minimum access necessary for the user to perform their duties. Regular reviews of role assignments should be conducted (recommended quarterly) to ensure that access remains appropriate as staff responsibilities change.

The role assignment process involves navigating to Administration > User Management, selecting the user, and adding or removing roles from the Role Assignments section. Changes take effect immediately upon saving. Users who are logged in will see their permissions update on their next page load.

F.3 Notification Configuration

The notification system sends alerts for key asset management events. Configure notification preferences to ensure the right people are informed at the right time without creating notification fatigue. The following notification types can be configured:

Asset Request Notifications: Sent to approvers when a new request requires their review. Configurable by approval stage and priority level.

Maintenance Due Notifications: Sent to asset custodians and maintenance teams when scheduled maintenance is approaching. Lead time is configurable (default: 7 days before due date).

Insurance Expiry Notifications: Sent to risk management when insurance policies are approaching renewal. Lead time is configurable (default: 30 days before expiry).

Depreciation Completion Notifications: Sent to the finance team when monthly depreciation batch processing is complete, including a summary of results and any exceptions.

Transfer Notifications: Sent to the receiving department head when an asset transfer is initiated to their department, requiring acknowledgement of receipt.

Disposal Approval Notifications: Sent to required approvers when a disposal request is submitted for authorisation.

Appendix G: Data Migration Guide

G.1 Overview

Migrating asset data from legacy systems (spreadsheets, previous software, or paper records) into the NCG Warehouse Solutions platform requires careful planning and execution. This guide provides a structured approach to data migration that minimises errors and ensures data integrity.

The migration process follows five phases: Assessment, Preparation, Transformation, Loading, and Verification. Each phase has specific deliverables and quality gates that must be met before proceeding to the next phase. A typical migration project takes 4-8 weeks depending on data volume and complexity.

G.2 Phase 1: Assessment

The assessment phase identifies the scope of data to be migrated and evaluates the quality of source data. Key activities include:

Source Data Inventory: Document all sources of asset data, including primary systems, spreadsheets, databases, and paper records. Identify the system of record for each data element.

Data Quality Assessment: Analyse source data for completeness, accuracy, consistency, and currency. Common issues include missing fields, duplicate records, inconsistent naming conventions, and outdated values.

Gap Analysis: Compare the source data structure with the NCG Warehouse Solutions schema to identify mapping gaps, required transformations, and data that must be sourced from elsewhere.

Volume Estimation: Count the total records to be migrated and estimate the time required for transformation and loading. Plan for multiple migration cycles (test loads before the final cutover).

G.3 Phase 2: Preparation

Preparation involves creating the migration plan, mapping documents, and transformation rules. Key deliverables include:

Field Mapping Document: A detailed mapping of every source field to its target field in the NCG system. Include data type conversions, default values for missing data, and validation rules.

Category Mapping: Map legacy asset categories to the new category hierarchy. This is often the most complex mapping as legacy systems frequently use inconsistent or ad-hoc classification schemes.

Location Mapping: Map legacy location codes to the new location hierarchy. Physical verification of locations may be required to resolve ambiguities.

Migration Script Development: Create automated scripts for data extraction, transformation, and loading. Manual migration is not recommended for more than 50 records due to error rates.

G.4 Phase 3: Transformation

Transformation converts source data into the format required by the target system. This includes data cleansing (correcting errors, standardising formats), enrichment (adding missing data from other sources), and structural transformation (mapping to target schema).

Common transformations include: standardising asset names to the [Brand] [Model] [Spec] convention, converting currency values to the base currency, calculating accumulated depreciation based on historical purchase dates and depreciation policies, and generating asset codes following the INV-{CAT}-XXX format.

Each transformation rule should be documented and validated against sample data before being applied to the full dataset. Edge cases and exceptions should be identified and handled explicitly rather than relying on default behaviour.

G.5 Phase 4: Loading

Loading inserts the transformed data into the production system. Best practices include:

Test Load: Perform at least two test loads into a non-production environment before the final migration. Verify record counts, data accuracy, and system functionality after each test load.

Sequence: Load reference data first (categories, locations, users), then master data (Asset Master records), then transactional data (individual assets, transfer history, depreciation records).

Batch Size: Load data in manageable batches (500-1000 records) to enable progress monitoring and easier error diagnosis. Do not attempt to load the entire dataset in a single transaction.

Cutover Window: Schedule the final production load during a maintenance window when users are not actively working in the system. Communicate the timeline to all stakeholders in advance.

G.6 Phase 5: Verification

Post-migration verification ensures that all data has been loaded correctly and the system is functioning as expected. Key verification activities include:

Record Count Reconciliation: Compare the number of records in the source system with the number loaded into the target system. Investigate any discrepancies.

Sample Data Verification: Select a random sample of migrated records (minimum 5% or 50 records, whichever is greater) and verify every field against the source data.

Financial Reconciliation: Compare total asset values, accumulated depreciation, and net book values between the source and target systems. These should match exactly.

Functional Testing: Execute key workflows (asset registration, transfer, depreciation run, report generation) using migrated data to verify system functionality.

User Acceptance: Have designated users from each department review their migrated data and confirm accuracy. Document any issues and corrections.

Appendix H: Document Change Log

Version	Date	Author	Changes
1.0	January 2026	System Documentation Team	Initial release covering core asset management features
1.1	February 2026	System Documentation Team	Added depreciation worked examples, expanded role matrices
1.5	March 2026	System Documentation Team	Added Mermaid process diagrams for all major workflows
2.0	April 2026	System Documentation Team	Major expansion: added maintenance, insurance, audit, CI/CD, integration chapters. Added configuration guide and data migration appendices. Expanded all chapters with UI walkthroughs and best practices. DOCX format.

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